

1 Kernels

1. Let $\chi = \mathbb{R}^+$

(a) Verify that $\min(x, y) = \int_0^\infty (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) dt$

Choose $x, y \in \mathbb{R}^+$.

Case 1, $x \geq y$

$$\begin{aligned} \int_0^\infty (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) dt &= \int_0^y (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) dt + \\ \int_y^\infty (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) &= \int_0^y (1)(1) dt + 0 = \int_0^y dt = y \end{aligned}$$

Case 2, $x \leq y$

$$\begin{aligned} \int_0^\infty (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) dt &= \int_0^x (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) dt + \\ \int_x^\infty (\mathbb{1}_{t < x})(\mathbb{1}_{t < y}) &= \int_0^x (1)(1) dt + 0 = \int_0^x dt = x \end{aligned}$$

Combining both cases gives the desired result.

(b) Show that $K(x, y) = \min(x, y)$ is pos. definite over $\mathbb{R}^+$.

Case 1, $x_i \leq x_j$ and $\alpha \in \mathbb{R}^n$

In the same manner as above

$$\begin{aligned} \sum_{i,j}^n \alpha_i \alpha_j \int_0^\infty (\mathbb{1}_{t < x_i})(\mathbb{1}_{t < x_j}) dt &= \int_0^{x_i} (\sum_i \alpha_i \mathbb{1}_{t < x_i})(\sum_j \alpha_j \mathbb{1}_{t < x_j}) dt = \\ \int_0^{x_i} (\sum_i \alpha_i)(\sum_j \alpha_j) dt &= (\sum_k \alpha_k)^2 x_i \geq 0 \text{ for } x_i \in \mathbb{R}^+. \end{aligned}$$

Likewise for $x_i \geq x_j$,

$$\begin{aligned} \sum_{i,j}^n \alpha_i \alpha_j \int_0^\infty (\mathbb{1}_{t < x_i})(\mathbb{1}_{t < x_j}) dt &= \int_0^{x_j} (\sum_i \alpha_i \mathbb{1}_{t < x_i})(\sum_j \alpha_j \mathbb{1}_{t < x_j}) dt = \\ \int_0^{x_j} (\sum_i \alpha_i)(\sum_j \alpha_j) dt &= (\sum_k \alpha_k)^2 x_j \geq 0 \text{ for } x_j \in \mathbb{R}^+. \end{aligned}$$

Hence K is pos. definite.

(c) Show that $\max(x, y)$ is not pos. definite over $\mathbb{R}^+$. Consider the case when $n=2$. Let

$v = (\alpha_1, \alpha_2)$ and $\gamma = \max(x_1, x_2)$. Then

$$\begin{aligned} v^T K v &= \begin{bmatrix} \alpha_1 & \alpha_2 \end{bmatrix} \begin{bmatrix} x_1 & \gamma \\ \gamma & x_2 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} \\ &= \alpha_1^2 x_1 + 2\alpha_1 \alpha_2 \gamma + \alpha_2^2 x_2 \end{aligned}$$

Since this must hold for any $v \in \mathbb{R}^2$, choose $\alpha_1 = -\alpha_2$.

Then the expression above becomes

$$\begin{aligned} \alpha_1^2 x_1 + 2\alpha_1 \alpha_2 \gamma + \alpha_2^2 x_2 &= \alpha_1^2 (x_1 + x_2 - 2\gamma) = \\ \alpha_1^2 (x_1 + x_2 - 2 \max(x_1, x_2)) &\leq 0 \end{aligned}$$

2. Let $(\Omega, \mathcal{A}, P)$ be a probability space

(a) Show that for any two events $A, B \in \mathcal{A}$, $K(A, B) = P(A \cap B)$ is pos. definite.

Choose $(\alpha_1, \dots, \alpha_n) \in \mathbb{R}^n$. Then

$$\begin{aligned} \sum_{ij}^n \alpha_i \alpha_j K(x_i, x_j) &= \sum_{ij}^n \alpha_i \alpha_j P(A_i \cap A_j) = \\ \sum_{ij}^n \alpha_i \alpha_j E(\mathbb{I}_{A_i} \mathbb{I}_{A_j}) &= E\left(\left(\sum_i \alpha_i \mathbb{I}_{A_i}\right) \left(\sum_j \alpha_j \mathbb{I}_{A_j}\right)\right) = \\ E\left(\left(\sum_k \alpha_k \mathbb{I}_{A_k}\right)^2\right) &\geq 0 \end{aligned}$$

(b) Show that $K(A, B) = P(A \cap B) - P(A)P(B)$ is also pos. definite.

Let $A = (A_1, A_2, \dots, A_n)$ be a vector of events.

$$\begin{aligned} \sum_{ij} \alpha_i \alpha_j (P(A \cap B) - P(A)P(B)) &= \sum_{ij} \alpha_i \alpha_j E(\mathbb{I}_{A \cup B}) = \\ \sum_{ij} \alpha_i \alpha_j (E(\mathbb{I}_{A_i} \mathbb{I}_{A_j}) - E(\mathbb{I}_{A_i})E(\mathbb{I}_{A_j})) &= \\ \sum_{ij} \alpha_i \alpha_j Cov(\mathbb{I}_{A_i}(\omega), \mathbb{I}_{A_j}(\omega)) \end{aligned}$$

where $\mathbb{I}_A(\omega) = 1$ if $\omega \in A$ and $\mathbb{I}_A(\omega) = 0$, otherwise. Furthermore

$$\begin{aligned} \sum_{ij} \alpha_i \alpha_j Cov(\mathbb{I}_{A_i}(\omega), \mathbb{I}_{A_j}(\omega)) &= \\ \sum \alpha_i^2 Var(\mathbb{I}_{A_i}(\omega)) + 2 \sum_{i,j|i < j} \alpha_i \alpha_j Cov(\mathbb{I}_{A_i}(\omega), \mathbb{I}_{A_j}(\omega)) &= \\ Var(\sum_i \alpha_i \mathbb{I}_{A_i}(\omega)) \geq 0. \end{aligned}$$

Credit: <https://en.wikipedia.org/wiki/Covariance>.

*Copied formula incorrectly first time.

3. $\mathcal{X} = \mathbb{N}^+$

(a) Show that $K(x, y) = LCM(x, y)$ is not pos. definite on $\mathbb{N}^+$.

Pick $\mathcal{X} = \{x_1, x_2\}$ where the x_i are distinct points in $\mathbb{N}^+$.

Choose $(\alpha_1, \alpha_2) \in \mathbb{R}^2$ with $\alpha_1 = -\alpha_2$.

Write $\{x_p\}_k = \{nx_k\}$ for $n \in \mathbb{N}$. Then

$$\begin{aligned} \sum_{ij}^n \alpha_i \alpha_j K(x_i, x_j) &= \sum_{ij}^n \alpha_i \alpha_j \min(\{x_p\}_i \cap \{x_p\}_j) = \\ &= \alpha_1^2 \min\{x_p\}_1 - 2\alpha_1^2 \min(\{x_p\}_1 \cap \{x_p\}_2) + \alpha_1^2 \min\{x_p\}_2 = \\ &= \alpha_1^2 (\min\{x_p\}_1 + \min\{x_p\}_2 - 2 \min(\{x_p\}_1 \cap \{x_p\}_2)) \leq 0 \end{aligned}$$

*had indices labeled i,j instead of 1,2 first time.

(b) Show that $K(x, y) = GCD(x, y)$ is pos. definite on $\mathbb{N}^+$.

Proof by Beslin and Ligh (1989): Let $X = (x_1, \dots, x_n) \in \mathbb{N}^{n+}$, $D = \{d_1, \dots, d_m\}$ be the set X union all the divisors of $x_i \in X$.

Let $\phi(x)$ denote the Euler's totient function. If we define a new $n \times m$ matrix $A = (a_{ij})$ and set

$$a_{ij} := e_{ij} \cdot \sqrt{\phi d_j}, \text{ with}$$

$$e_{ij} = \{1 \text{ if } d_j | x_i, 0 \text{ otherwise}$$

Then we have

$$(AA^T)_{ij} = \sum_k a_{ik}a_{jk} = \sum_{d_k | x_i, d_k | x_j} \sqrt{\phi(d_k)}\sqrt{\phi(d_k)} = \sum_{d_k | \gcd(x_i, x_j)} \phi(d_k) = \gcd(x_i, x_j)$$

The last equality is a known property of the totient function.

Since any matrix that can be decomposed into the product of another matrix with its transpose is pos. definite, we can conclude that $K(x, y)$ is as well.

4. let $x, y \in \mathbb{R}$ and let $k_\sigma(x, y) = \exp\left(-\frac{(x-y)^2}{2\sigma^2}\right)$

(a) For $\sigma = 1$. Prove that $K(x, y)$ is pos.definite using the the characteristic function.

Since $E[e^{itZ}] = e^{-t^2/2}$ we can set $x_i - x_j = t$ and notice the characteristic function is symmetric with respect to x_i, x_j . Then for $(\alpha_1 \dots \alpha_n) \in \mathbb{R}^n$

$$\sum_{ij} \alpha_i \alpha_j k(x_i, x_j) = \sum_{ij} \alpha_i \alpha_j E(e^{(x_i - x_j)Z}) = E\left(\sum_{ij} \alpha_i \alpha_j (e^{iZx_i} e^{-iZx_j})\right) = E\left(\sum_i \alpha_i e^{iZx_i} \sum_j \alpha_j e^{-iZx_j}\right) = E\left(\sum_k \alpha_k e^{iZx_k}\right)^2 \geq 0$$

The last equality follows from the symmetry of K .

(b) Show that k_σ is pos. definite for any $\sigma > 0$

Set $t = (x_i - x_j)/\sigma$, then following part (a) exactly

$$E\left(\sum_i \alpha_i e^{iZx_i/\sigma} \sum_j \alpha_j e^{-iZx_j/\sigma}\right) = E\left(\sum_k \alpha_k e^{iZx_k/\sigma}\right)^2 \geq 0$$

*added missing α_i 's from the first version